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Student Name: MD Samim Rain

London Met ID: 23050299

College ID: np02cp4a230163

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Submitted To: Mr. Binod Bhattarai

GitHub Link	
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I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

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1. Title of the Project

Comparison the study of Machine Learning Models for Predicting Pharmaceutical Drug Spending Using State Drug Utilization Data.

2. Introduction

Medicine in increasing many countries especially United State of America. Governments and health organizations spend a lot of money on medicines for public health programs. Because of this, it is important to understand how drug spending changes and what factors affect it. By studying drug spending data, I can find useful patterns and trends that help in better planning and decision-making.

In this project, a pharmaceutical drug spending dataset is used to analyses and predict drug costs. The dataset is taken from public sources provided by the U.S. government and is available on the Hugging Face platform. This dataset includes information such as state-wise drug utilization, total spending, and number of prescriptions. Machine learning is helpful in this case because it can learn from past data and make predictions about future drug spending (CMS.gov, n.d.).

In this coursework I used three model which I predict the pharmaceutical drug spending using ML. Therefore, Linear Regression, Random Forest, and Ensemble Model I send which is aimed to predict the about medicine and compassion. When I compare these three models then I confirm which model is more accurately given the output best result for medicines predictions. Now this model shows how artificial intelligence technology cab be applied to the real-world data predicted.

3. Background of the study

Drug spending is very high in many countries. Every year, governments and people spend a lot of money to buy medicines. If we can know how much money is spent and which states spend more, it will be easy to plan and save money. Many data about drug use and cost is now available from government websites. Machine learning can help to study this data and predict future spending (pharmapub, n.d.). Using these predictions, governments and hospitals can make better decisions about budgets and medicine

supply. Some studies also show that machine learning models like Linear Regression, Random Forest, and Ensemble are useful to predict healthcare costs (Link, 2024).

4. Related Works

Many researchers use machine learning to predict healthcare and drug costs. One study used Linear Regression and Random Forest to predict medicine prices. They found that Random Forest gave better results than Linear Regression. Another study used forecasting models to predict future drug spending in pharmacies. It showed that predictions can help plan budgets and resources. These studies show that machine learning is useful to understand spending patterns and predict costs. This project is similar because it uses state drug data and models to predict spending.

5. Problem Statement

The cost of medicine is very high now. Government and people spend a lot of money for drugs every year. It is hard to know how much money will be needed in the future. If we can predict drug spending, then government and hospitals can plan their money and budget better. Machine learning can help to do this because it can learn from old data and give prediction for future. This project will use Linear Regression, Random Forest, and Ensemble models to predict drug spending in different states.

6. Problem Solving

To solve the problem of high drug spending, we can use machine learning models. First, we will collect the state drug data from the government. Then we will use Linear Regression, Random Forest, and Ensemble models to predict future drug spending. These models will help us know which states spend more money and how much money is needed. By using this prediction, government and hospitals can plan their budget better. This will save money and help to provide medicine to more people.

7. Objective

To know how much money is spent on drugs in different states.

To use machine learning models to predict future drug spending.

To compare Linear Regression, Random Forest, and Ensemble models.

To help government and hospitals plan budget for medicine.

To save money and provide medicine to more people.

8. Data Description

The dataset is from HHS Official Dataset 2024. It has information about drug spending in different states of the United States. The data shows how much money is spent on each drug and how many prescriptions are given. It also has the name of the state and the type of drug. This data is useful to see which state spends more and to predict future spending using machine learning models. The dataset is available on Hugging Face website.

- **Dataset Source:** HHS Official Dataset 2024, Hugging Face.
- **Type:** Tabular data (Number and text)
- **Instances:** Many rows of data (each row shows drug spending for a state).
- **Features:** State name, Drug name, Number of prescriptions, Total spending, Type of drug.
- **Target Variable:** Total spending of drugs.

Missing values: Some values are missing in few rows, need to clean before using.

9. Dataset Visualization.

9.1. Histogram of Total Spending

This graph shows how much money each state spends on drugs. Some states spend more money and some states spend less. It helps to see which states spend a lot and which states spend less.

9.2. Graph of Number of Prescriptions

This graph shows how many prescriptions are given for each drug in each state. Some drugs have many prescriptions and some have few. It helps to know which drugs are used more.

9.3. Heatmap of Features

This graph shows the connection between features like number of prescriptions and total spending. Dark colour means strong connection and light colour means weak connection. It helps to know which factors affect spending more.

9.4. Bar Chart of Top 10 Drugs by Spending

This graph shows the 10 drugs with the highest spending. It helps to know which drugs cost more money.

9.5. Pie Chart of Spending by State

This graph shows how money is spent in each state as a part of total spending. It helps to see which state uses more money for drugs.

9.6. Line Graph of Spending Over Time

This graph shows how spending changes over time. It helps to see if spending is increasing or decreasing.

9.7. Box Plot of Total Spending

This graph shows the spread of total spending and if there are any very high or very low values. It helps to find unusual values in data.

10. Proposed ML Algorithms

In this project, we will use three machine learning models to predict drug spending:

10.1. Linear Regression

This model is simple. It tries to find a line that can show the relation between features like number of prescriptions and total spending. It gives quick results but may not work well for complex data.

Pseudocode:

Start

Load the state drug utilization dataset from Hugging Face

Clean the dataset and handle missing values

Select relevant input features (State, Drug Name, Number of Prescriptions, Drug Type)

Select target variable (Total Spending)

Split the dataset into training and testing sets

Train Linear Regression model on training data

Use the trained Linear Regression model to predict drug spending on testing data

Calculate prediction accuracy

End

10.2. Random Forest

This model uses many decision trees and combines their results. It can find patterns in complex data and usually gives better results than Linear Regression.

Pseudocode:

Start

Load the state drug utilization dataset from Hugging Face

Clean the dataset and handle missing values

Select relevant input features (State, Drug Name, Number of Prescriptions, Drug Type)

Select target variable (Total Spending)

Split the dataset into training and testing sets

Create many decision trees using training data

Train the Random Forest model

Use the trained Random Forest model to predict drug spending on testing data

Calculate prediction accuracy

End

10.3. Ensemble Model

This model combines two or more models, like Linear Regression and Random Forest, to make better predictions. It reduces mistakes from single models and improves accuracy.

Pseudocode:

Start

Load the state drug utilization dataset from Hugging Face

Clean the dataset and handle missing values

Select relevant input features (State, Drug Name, Number of Prescriptions, Drug Type)

Select target variable (Total Spending)

Split the dataset into training and testing sets

Train the Linear Regression model using training data

Train the Random Forest model using training data

Predict drug spending using Linear Regression on testing data

Predict drug spending using Random Forest on testing data

Combine predictions from both models

Generate final prediction using Ensemble model

Calculate prediction accuracy

End

11. Methodology

First, the drug spending dataset is collected from the Hugging Face website. Then the data is cleaned by removing missing and wrong values. After that, useful features like state name, drug name, number of prescriptions, and drug type are selected. The target value is total drug spending. The dataset is divided into training data and testing data. Linear Regression, Random Forest, and Ensemble models are trained using the training

data. These models are then used to predict drug spending on testing data. Finally, the results of all models are compared to find which model gives better prediction.

12. Evaluation Metrics

In this project, we calculate some values to check how well the models work. We calculate Mean Absolute Error (MAE) to know the average difference between predicted spending and real spending. We also calculate Mean Squared Error (MSE) to know how big the prediction error is. Another value is R-squared, which shows how much the model fits the data. Smaller MAE and MSE values mean the prediction is better. These calculations help us compare Linear Regression, Random Forest, and Ensemble models and choose the best one.

13. Expected outcome

After completing this project, we expect that the machine learning models will predict drug spending correctly. Random Forest and Ensemble models are expected to give better results than Linear Regression. The project will help to know which states spend more money on drugs. It will also help government and hospitals to plan their budget in a better way. This project shows that machine learning can be useful in healthcare data analysis.

14. Technology and Tools

- VS Code – used to write and run the program
- Python – used as the main programming language
- NumPy – used for numerical calculation
- Pandas – used to load and clean the dataset
- Scikit-learn – used to apply Linear Regression, Random Forest, and Ensemble models
- Matplotlib – used to create graphs and charts
- Hugging Face – used as the dataset source
- Jupyter Notebook - tool used to write and run Python code.

15. Conclusion

In this project, we used machine learning to predict drug spending in different states. Linear Regression, Random Forest, and Ensemble models are used to see which model gives better results. Random Forest and Ensemble models give better prediction than Linear Regression. This project helps to know which states spend more money on drugs. It also helps government and hospitals to plan their budget and save money. Machine learning is very useful for predicting healthcare costs and understanding drug spending patterns.

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